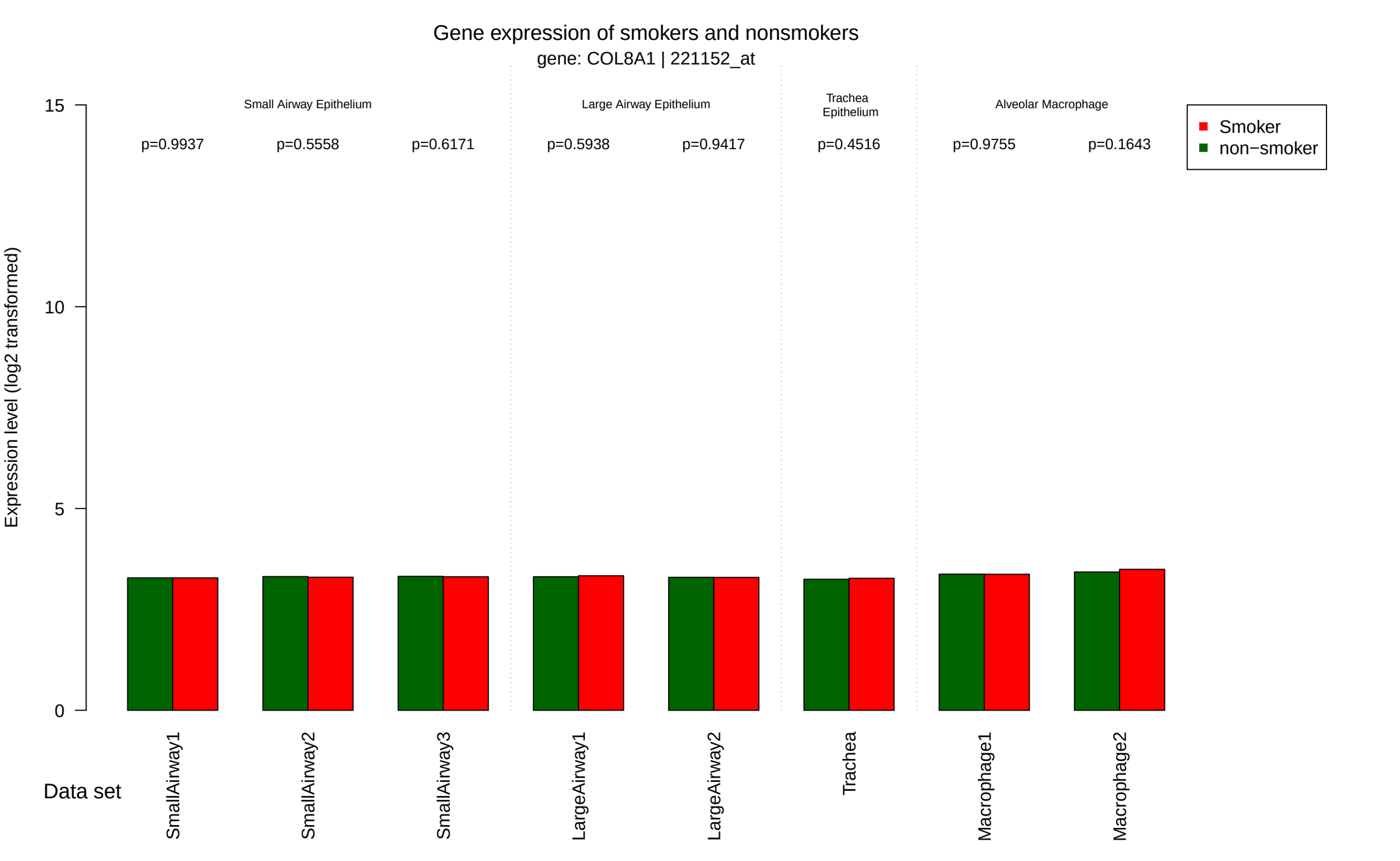
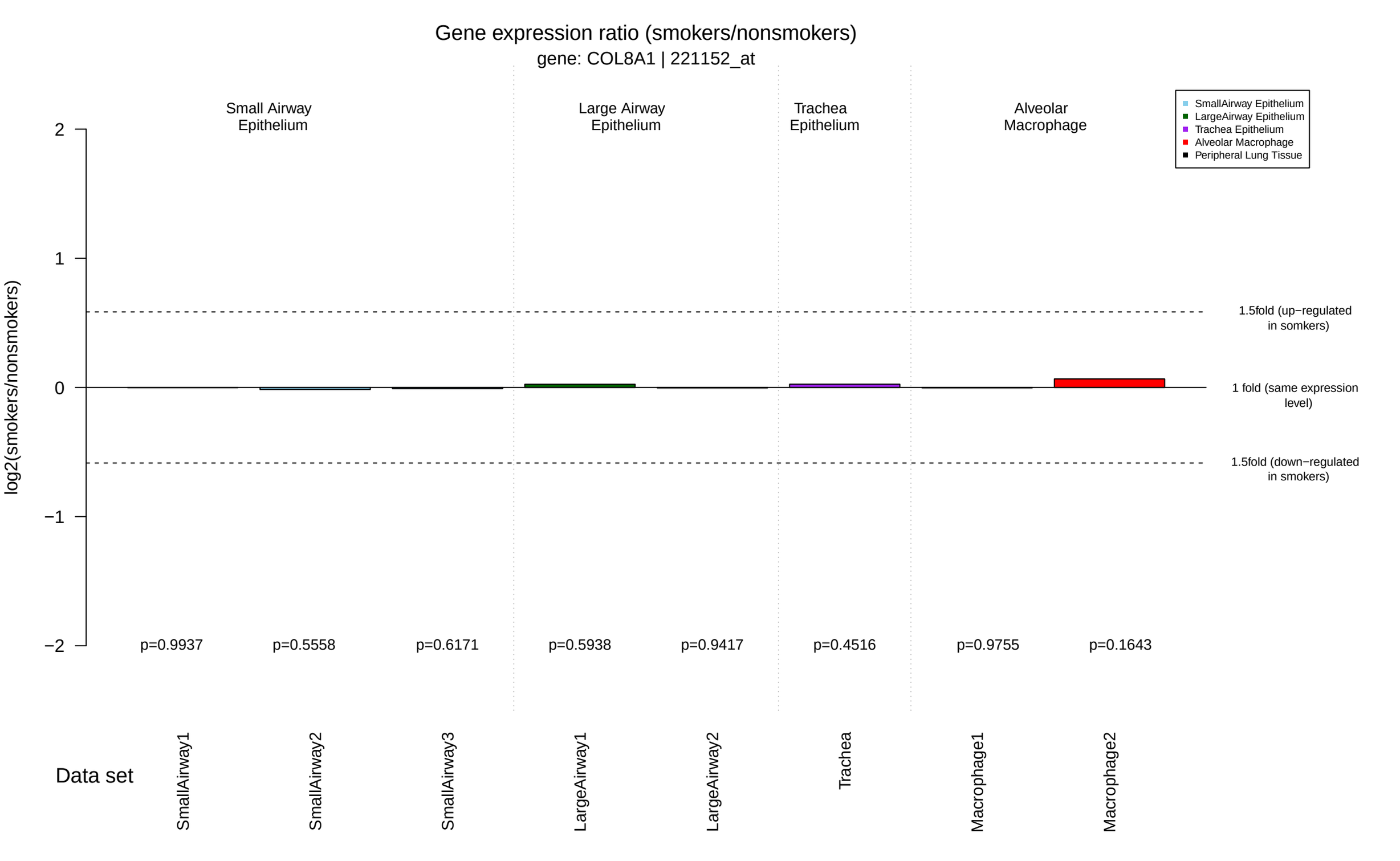
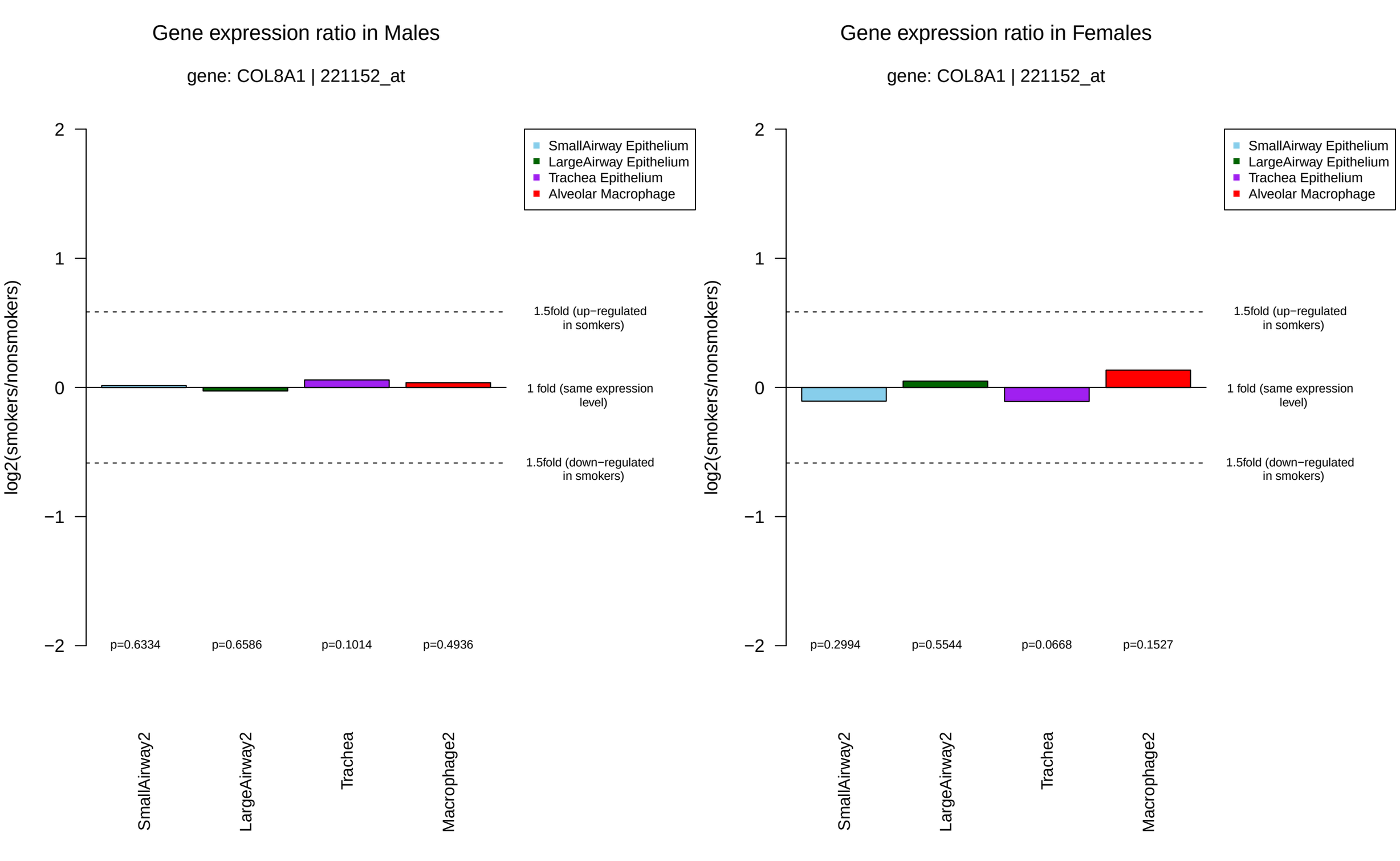
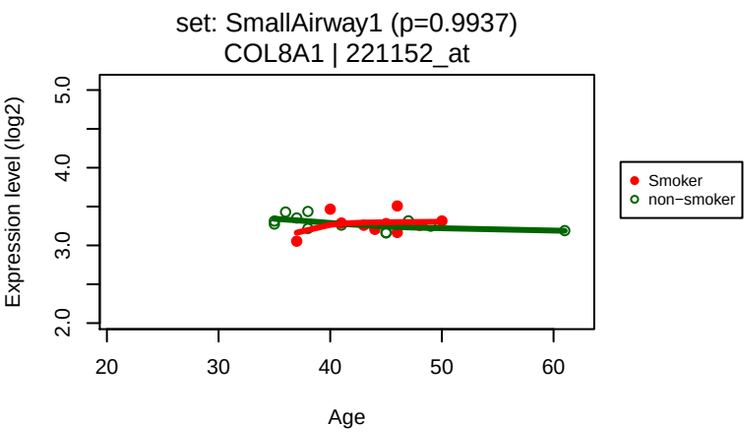
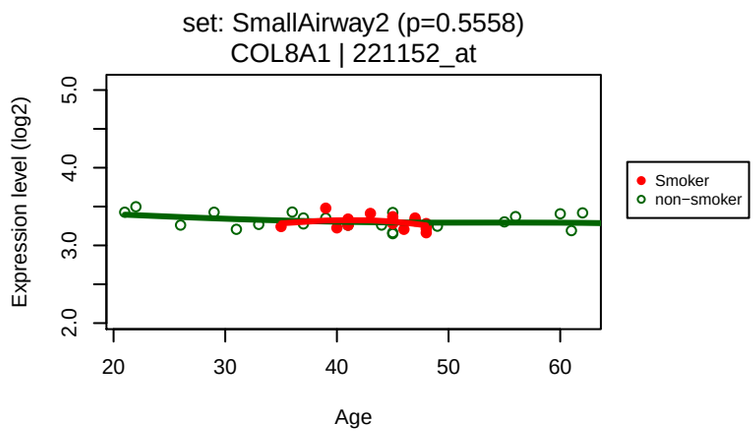
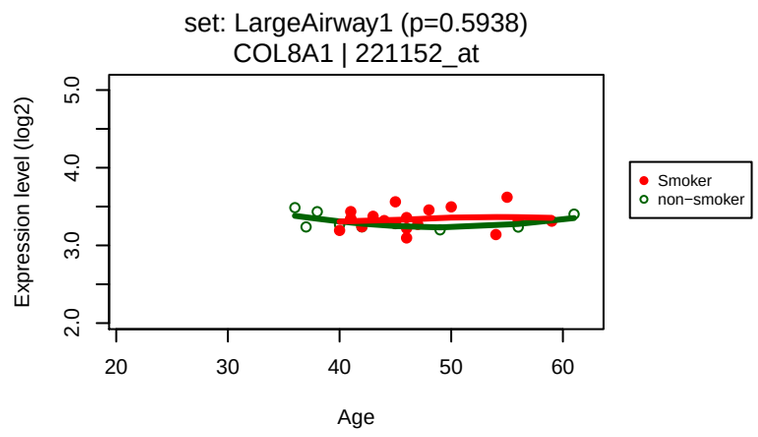

Gene expression of smokers and nonsmokers
gene: COL8A1 | 221152_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9937
p=0.5558
p=0.6171
p=0.5938
p=0.9417
p=0.4516
p=0.9755
p=0.1643
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

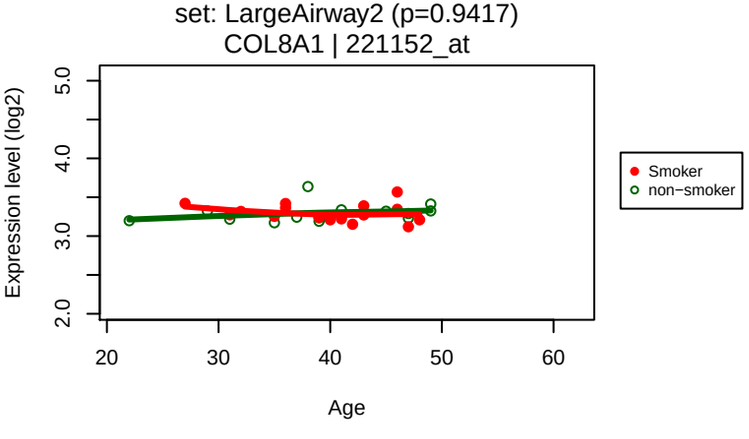

Gene expression ratio (smokers/nonsmokers)
gene: COL8A1 | 221152_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9937
p=0.5558
p=0.6171
p=0.5938
p=0.9417
p=0.4516
p=0.9755
p=0.1643
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

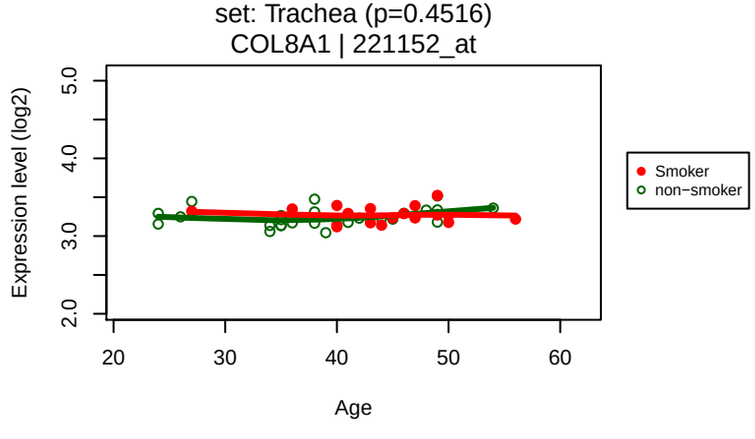

Gene expression ratio in Males
gene: COL8A1 | 221152_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6334
p=0.6586
p=0.1014
p=0.4936
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL8A1 | 221152_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2994
p=0.5544
p=0.0668
p=0.1527
SmallAirway2
LargeAirway2
Trachea
Macrophage2

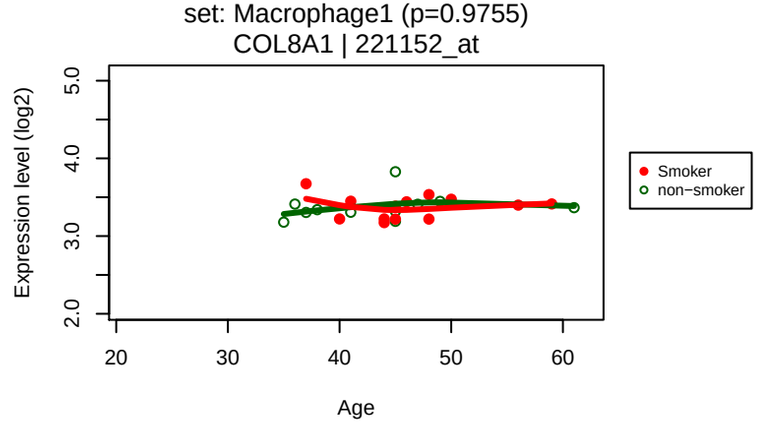

set: SmallAirway1 (p=0.9937)
COL8A1 | 221152_at
Expression level (log2)
Age
Smoker
non−smoker

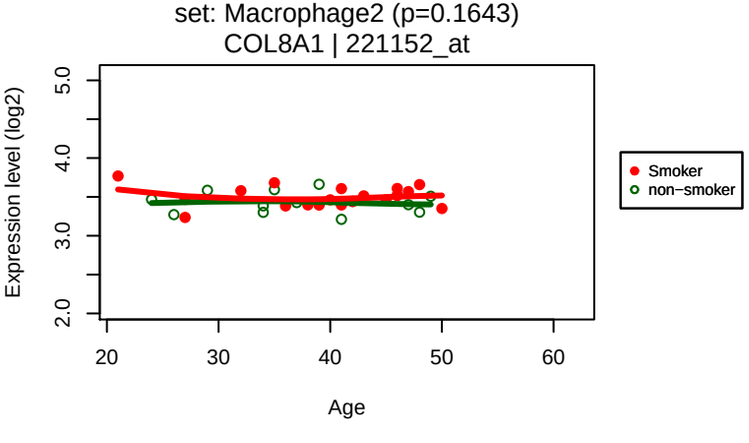

set: SmallAirway2 (p=0.5558)
COL8A1 | 221152_at
Expression level (log2)
Age
Smoker
non−smoker

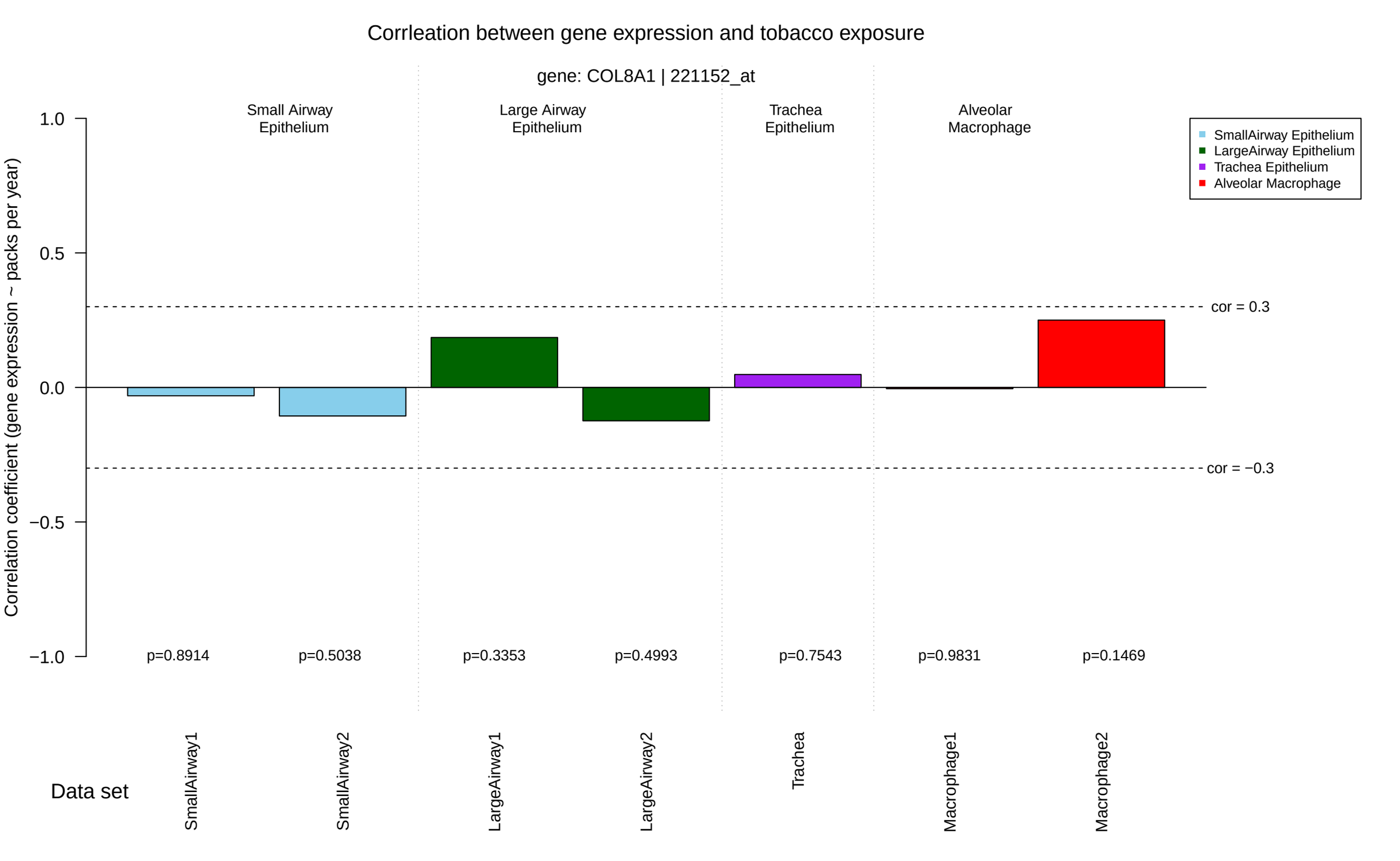

set: LargeAirway1 (p=0.5938)
COL8A1 | 221152_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9417)
COL8A1 | 221152_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4516)
COL8A1 | 221152_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9755)
COL8A1 | 221152_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1643)
COL8A1 | 221152_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL8A1 | 221152_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8914
p=0.5038
p=0.3353
p=0.4993
p=0.7543
p=0.9831
p=0.1469
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2